LOKOGAD: Design of a Low-Cost Rehabilitation Exoskeleton Based on a Modified Theo Jansen Mechanism for Human Bipedalism via the Graph-Assisted Design Framework

Gabriel A. Laurente Caldas
Faculty of Biomedical Engineering
Universidad Nacional Mayor de San Marcos
Lima, Peru
gabriel.laurente@unmsm.edu.pe

Axel T. Mora Zeña
Faculty of Biomedical Engineering
Universidad Nacional Mayor de San Marcos
Lima, Peru
axel.mora@unmsm.edu.pe

Diego F. Pantoja Iturrizaga
Faculty of Biomedical Engineering
Universidad Nacional Mayor de San Marcos
Lima, Peru
diego.pantoja@unmsm.edu.pe

Jimmy F. León Vásquez
Faculty of Biomedical Engineering
Universidad Nacional Mayor de San Marcos
Lima, Peru
jimmy.leonv@unmsm.edu.pe

***Abstract*—Commercial rehabilitation exoskeletons cost between 40,000 and 100,000 USD, which sharply limits access to gait therapy for low-income populations. This paper presents LOKOGAD, a low-cost rehabilitation exoskeleton concept based on a biomimetic modification of the Theo Jansen leg mechanism, adapted to human bipedalism through a Graph-Assisted Design (GAD) framework. The scope of this work is kinematic: the goal is to replicate standard human gait trajectories, and forces, torques, and tests with human subjects are not addressed. The GAD framework abstracts the mechanism as a computational graph, combining exact analytical kinematics with geometric-constraint validation. The classic Jansen linkage, originally conceived for quadrupedal locomotion, is reconfigured through multi-objective evolutionary optimization (NSGA-II) followed by a topological refinement stage, using reference trajectories derived from the OpenSim Gait2392 model. The optimized mechanism achieves a Procrustes error of 4.97 percent in foot trajectory and 10.79 percent in knee trajectory, with a total horizontal foot excursion of 966.35 mm at human scale. A physical prototype fabricated by fused deposition modeling and driven by a single generic DC motor shows that the optimized design can be assembled and set in motion. Because the system has a single rotary input and a purely mechanical transmission, it requires no per-joint actuators or drivers, which in principle makes it a less expensive architecture than multi-axis exoskeletons.**

***Index Terms*—additive manufacturing, gait rehabilitation, graph-assisted design, multi-objective optimization, NSGA-II, rehabilitation exoskeleton, Theo Jansen mechanism**

I. Introduction

Cerebrovascular disease, spinal cord injuries, and degenerative neurological disorders affect over 15 million people annually worldwide, many of whom require gait rehabilitation therapy to recover mobility. Robotic assistive devices, generically known as lokomats, have proven effective at accelerating functional recovery through robot-assisted gait therapy. However, available commercial systems—such as Lokomat (Hocoma), EksoNR (Ekso Bionics), and ReWalk (ReWalk Robotics)—cost between $40,000 and $100,000 USD, placing them out of reach for most public health centers and patients in developing countries. Low- and middle-income countries account for roughly 80% of the world's population with motor disabilities, yet hold less than 10% of globally available rehabilitation resources, making low-cost, manufacturable rehabilitation exoskeletons both a technological and social imperative.

This paper presents LOKOGAD, a lower-limb rehabilitation exoskeleton built on three pillars. First, a **single-degree-of-freedom, low-cost mechanism**: the Theo Jansen gait mechanism, composed of rigid links coupled in closed kinematic loops and driven by a single motor, which reduces the number of actuators, drivers, and control channels (and, in principle, the cost) relative to traditional exoskeletons requiring independent actuation at each joint (hip, knee, ankle). Second, **biomimetic optimization via the GAD framework**, which abstracts the mechanism as a computational graph topology, enabling multi-objective evolutionary optimization (NSGA-II) to adapt the Jansen kinematics to bipedal human gait biomechanics. Third, **accessible additive manufacturing**, with the full mechanical assembly designed for FDM 3D printing in low-cost materials (PLA/PETG) through an automated GAD pipeline that generates collision-free, print-ready 3D models.

Linkage synthesis and computational mechanism design are well-studied fields. Deb et al. [1] introduced NSGA-II as a computationally efficient multi-objective optimizer, which has become a standard tool for evolutionary approaches in

mechanical design, while Liao and McCarthy [2] solved the seven-position synthesis of a spatial 5-SS platform linkage, an example of exact finite-position linkage synthesis. Graph-based representations have also been used for automatic design: Wang et al. [3] formulated robot design as a search over graphs and solved it with neural graph evolution. In low-cost exoskeletons, Slucock [4] systematically reviewed low-cost actuator implementations for lower-limb exoskeletons from a technical and financial perspective, and Narayan and Dwivedy [5] designed a low-cost lower-limb exoskeleton for paediatric gait rehabilitation. The Theo Jansen mechanism itself has been studied dynamically by Nansai et al. [6], and a modified Jansen linkage with fewer links was proposed and compared kinematically by Panchasara et al. [7], but, to the authors' knowledge, no prior work has integrated a Jansen-mechanism modification for human bipedalism with the GAD framework as a low-cost rehabilitation exoskeleton.

This paper's main contributions are: (1) the design of LOKOGAD, a low-cost rehabilitation exoskeleton based on a biomimetic modification of the Theo Jansen mechanism for human bipedalism, driven by a single rotary input through a purely mechanical transmission; (2) the GAD framework for graph-based abstraction and optimization of the mechanism, integrating exact analytical kinematics with robust geometric-constraint validation and NSGA-II multi-objective optimization; (3) a physical prototype fabricated via FDM additive manufacturing, which was assembled and set in motion, showing that the optimized design can be built and moved (assembly and motion capability; trajectory accuracy was not measured on the prototype); (4) biomimetic optimization achieving a 4.97% Procrustes error in foot trajectory against reference trajectories derived from the OpenSim Gait2392 model; and (5) an automated pipeline for collision-free 3D-model generation with graph-coloring-based Z-layer assignment for FDM manufacturing.

The contribution of this paper is kinematics-only: none of the optimization objectives includes a force, torque, or load-bearing criterion, and the prototype was not tested with a patient or under load. These limitations respond to the resources available for this stage of the project.

II. Theoretical Framework

A. Theo Jansen Mechanism and Its Adaptation to Bipedalism

The Theo Jansen mechanism is a planar linkage composed of 8 revolute joints and 15 connecting segments. Originally conceived for quadrupedal locomotion over sandy terrain, it is driven by a single rotary crank input, producing a cyclic foot trajectory. Its 13 variable bar lengths—commonly termed "Magic Numbers"—form the design genome that determines the output trajectory's kinematic characteristics. For adaptation to human bipedalism, the Jansen mechanism offers key advantages for rehabilitation: a single degree of freedom that simplifies control and reduces cost, an inherently stable cycloidal trajectory during stance, and the ability to generate both swing and stance phases with one actuator. However, Jansen's original proportions produce a gait that is biomechanically inefficient for human bipedalism, motivating optimization through the GAD framework.

B. Procrustes Analysis for Biomechanical Evaluation

Procrustes analysis is a standard statistical method for comparing shapes after removing translation, scale, and rotation effects. Given two trajectories X and Y, the method centers both at the origin by subtracting their centroids, normalizes both to unit scale, and finds the optimal rotation matrix R via singular value decomposition (SVD) that minimizes the root-mean-square deviation,

$$R = UV^T, \quad \text{where} \quad USV^T = \mathrm{SVD}(Y^T X). \tag{1}$$

The Procrustes error is then computed as

$$\epsilon_P = \frac{1}{N} \sum_{i=1}^{N} \|x_i - R y_i\|^2. \tag{2}$$

A value of $\epsilon_P = 0$ indicates identical shapes, while $\epsilon_P = 1$ indicates the absence of correlation. This error constitutes the primary metric for evaluating the exoskeleton's biomechanical fidelity relative to reference human gait.

C. NSGA-II Multi-Objective Optimization

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) [1] is a population-based evolutionary algorithm designed for multi-objective optimization problems. It maintains diversity via crowding distance and uses non-dominated sorting to rank solutions, employing Simulated Binary Crossover (SBX) and Polynomial Mutation (PM) operators to explore the design space efficiently.

III. Methodology

The GAD pipeline comprises five interconnected phases that progressively transform reference biomechanical data into the final optimized, manufacturable exoskeleton. Fig. 1 illustrates the complete LOKOGAD workflow.

A. Phase 1: Trajectory Extraction and Preprocessing

Reference biomechanical data were extracted from the Gait2392 musculoskeletal model, developed by Stanford University's OpenSim biomechanics laboratories [8]. Joint trajectories (foot, knee, hip, ankle) were exported as a MATLAB structure containing 73 3D spatial coordinates over a complete normalized gait cycle (0% to 100%), loaded via `scipy.io.loadmat()` and cubically interpolated from 73 frames to 360 points, corresponding to 1° increments of the crank rotation cycle. Raw trajectories contain high-frequency stochastic noise and aliasing artifacts from numerical integration; to obtain mathematically continuous curves, the discrete kinematics are transformed via truncated Fourier series,

$$f(\theta) = a_0 + \sum_{k=1}^{H} \left[a_k \cos(k\theta) + b_k \sin(k\theta) \right], \tag{3}$$

using a weighted loss $\mathcal{L} = w_{MSE}\mathcal{L}_{MSE} + w_S\mathcal{L}_{Smooth} + w_{L2}\mathcal{L}_{L2}$, where $\mathcal{L}_{MSE}$ prioritizes the X and Z axes (stride motion), $\mathcal{L}_{Smooth}$ penalizes large second derivatives, and $\mathcal{L}_{L2}$ regularizes high harmonic amplitudes ($k > 4$).

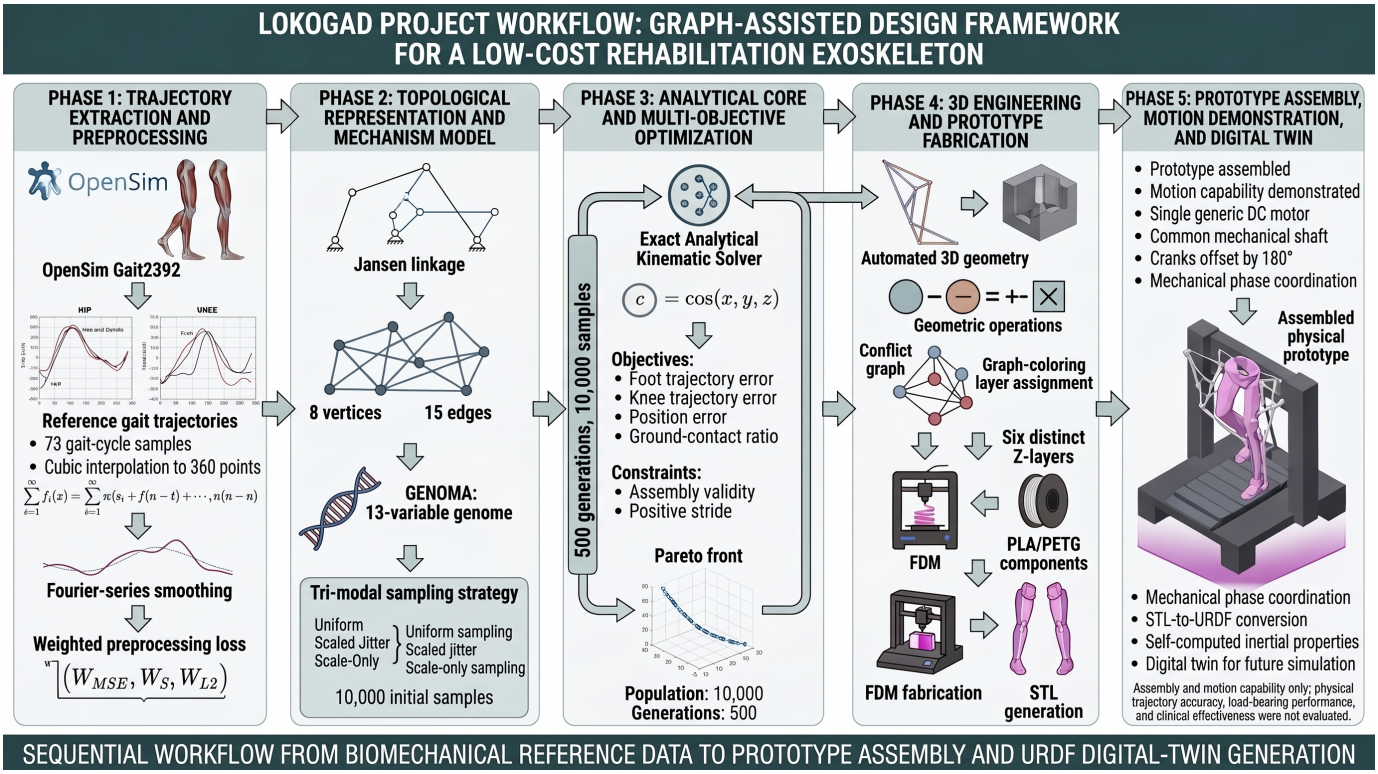


Fig. 1. LOKOGAD project workflow (GAD Framework) for a low-cost rehabilitation exoskeleton. The pipeline comprises five phases: (1) extraction and preprocessing of biomechanical trajectories from OpenSim, (2) topological representation of the mechanism as a graph with 13 Magic Numbers, (3) analytical core and NSGA-II multi-objective optimization, (4) automated 3D engineering and FDM prototype fabrication, and (5) prototype assembly, motion demonstration, mechanical phase coordination, and URDF digital-twin generation.

B. Phase 2: Topological Representation and Sampling

The Jansen mechanism is abstracted as a planar graph: 8 pivot vertices carry feature vectors indicating semantic type (fixed anchor, crank-driven, foot end-effector); 15 edges encode normalized geometric bar lengths; and the genome of the parametric stage consists of 13 floating-point values, the "Magic Numbers." (The topological stage described in Phase 3 operates on the full graph instead.) A design-space dataset of 10,000 samples was generated using a tri-modal sampling strategy: Uniform (40%, exploration over $[0.3\times, 40\times]$ the original genome), Scaled Jitter (40%, genome scaled $5\times$–$25\times$ with $\pm 30\%$ Gaussian perturbation), and Scale-Only (20%, uniform scaling $1\times$–$30\times$ without perturbation).

C. Phase 3: Algebraic Core and Multi-Objective Optimization

The GAD computational core implements an exact analytical vector solver for circle–circle intersection. Given bases P_1 with radius r_1 and P_2 with radius r_2, the algorithm determines the intersection points; critical heuristics handle degenerate cases (non-intersecting or contained circles) by returning NaN indicators, flagging the mechanism as "Assembly Failed." The optimization problem has 4 minimization objectives: **foot_error** and **knee_error** (Procrustes error vs. OpenSim), $|\text{ground_ratio} - 0.55|$ (deviation from the ideal 55% ground-contact ratio), and **pos_error** (absolute centroid position error); plus 2 hard constraints: **G1** (assembly validity—any NaN among the 360 angles is penalized with $F = [10.0, 10.0, 1.0, 10.0]$) and **G2** (positive stride—stride_mm > 0). Configuration: population = 10,000, generations = 500, operators = SBX + PM, parallelized with `multiprocessing.Pool`.

Topological refinement. The parametric search is restricted to the fixed Jansen topology (13 lengths). To relax this restriction, a second evolutionary stage operates directly on the graph over 500 generations, with tournament selection, elitism, a length-interpolating crossover, and mutations that perturb bar lengths and anchor positions, add or remove pivots and bars, and reassign the foot and knee outputs. Individuals are ranked by a scalar *combined fitness*: $F = 0.40\,\epsilon_{foot} + 0.25\,\epsilon_{knee} + 0.30\,\epsilon_{pos} + 0.05\max(0, |s-1000|/1000) + 0.001\max(0, N_b - 12)$, where ϵ_{foot}, ϵ_{knee} are the Procrustes errors, $\epsilon_{pos} = 0.6\,\epsilon_{pos,foot} + 0.4\,\epsilon_{pos,knee}$ is the absolute position error, s is the stride in mm, and N_b is the number of bars of the mechanism (the penalty applies only above 12 bars, so it is zero for the selected 12-bar individual). The winning individual of this stage has 8 pivots and 12 bars; it is the mechanism evaluated in Table III and Fig. 4 and the one that was fabricated.

D. Phase 4–5: Fabrication and Digital Twin

Each bar is modeled in Shapely as the convex buffer of its endpoint circles, with hole

subtraction accounting for FDM print tolerance (PRINT_TOL = 0.4 mm radial clearance), extruded in Z via `trimesh.creation.extrude_polygon()`. For collision-free Z-layer assignment, bar pairs are tested across all 360 simulation frames for intersections using cross-product determinants; conflicts are recorded in a "Conflict Graph," and NetworkX greedy graph coloring assigns integer labels ensuring no conflicting bars share the same Z-layer, with an inter-layer pitch of BAR_THICK $\times$ 2.2. The physical prototype was fabricated via FDM 3D printing in PLA and PETG: Jansen-mechanism bars were printed in white, while biped-support structural parts were printed in magenta for visual identification. Assembly uses M3 screws and standard ball bearings for revolute joints. The two mechanisms (one per limb) are mounted on a support structure and driven by a single generic DC motor (of the kind used in remote-control cars) through a common shaft, with the two cranks offset by 180°; this sets the phase between the limbs mechanically. The motor is only a source of rotary motion, not a torque-providing actuator. STL meshes are validated and transpiled to URDF with self-computed inertial properties, enabling future development of reinforcement-learning gait controllers.

IV. Results and Discussion

A. Optimization Convergence History

NSGA-II was run for 500 generations with a population of 10,000 individuals. Fig. 2 shows the convergence history across four metrics. The sharpest improvement occurred between generations 1 and 23, where the best individual's *foot error* dropped from 0.2205 to 0.0457, a 79.3% reduction. From this point on, the tracked metric is the *combined fitness*: between generations 23–100 its curve flattened significantly, indicating convergence toward the main Pareto valley, and generations 100–500 constituted only microscopic refinement (0.11848 $\rightarrow$ 0.11177, a further 5.7% improvement). The final foot error (0.0497) is slightly above its generation-23 value because the combined fitness also weights the knee and position errors. The number of valid solutions grew from 2,422 (24.2%) at generation 1 to 3,941 (39.4%) at generation 500.

B. Optimized Genomes

Table I presents the 13 optimized Magic Numbers of the parametric stage (Stage 1) compared with Jansen's original proportions; Fig. 3 illustrates the comparison and per-bar scale factor. The average scale factor is approximately $\times$18.3 with respect to the original Jansen proportions; it is a scaling to human gait dimensions and does not refer to the print bed. Table II gives the bar lengths of the final 12-bar mechanism obtained by topological refinement, which is the mechanism evaluated in Table III and fabricated; it is a different mechanism from the one in Table I. Table II lists the human-scale lengths used by the optimizer and the prototype lengths, obtained by a uniform downscaling of about 1:5.8 ($\times$0.1713). All links were printed as separate STL parts; the longest printed link (bar 9) has a center-to-center length of 114.3 mm and therefore fits comfortably in a 250$\times$250 mm print bed.

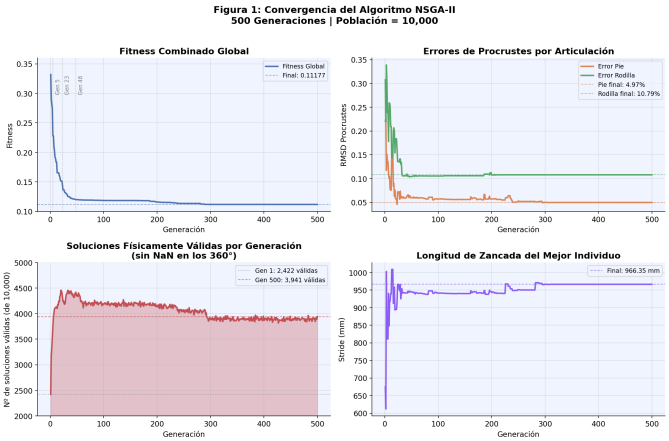


Fig. 2. NSGA-II convergence over 500 generations with a population of 10,000. Shown: global combined fitness (final: 0.11177), per-joint Procrustes error (final foot: 4.97%, knee: 10.79%), physically valid solutions per generation, and best-individual stride length (966.35 mm).

TABLE I
Optimized Genome of the Parametric Stage: The 13 Magic Numbers

ID	Optimized (mm)	Original (mm)	Scale Factor
1	15.55	38.0	$\times$0.41
2	224.97	7.8	$\times$28.84
3	102.43	15.0	$\times$6.83
4	1150.85	50.0	$\times$23.02
5	1272.60	41.5	$\times$30.67
6	817.33	39.3	$\times$20.80
7	919.51	61.9	$\times$14.85
8	389.20	55.8	$\times$6.97
9	1018.72	40.1	$\times$25.40
10	1188.35	39.4	$\times$30.16
11	706.39	36.7	$\times$19.25
12	651.60	65.7	$\times$9.92
13	1035.31	49.0	$\times$21.13

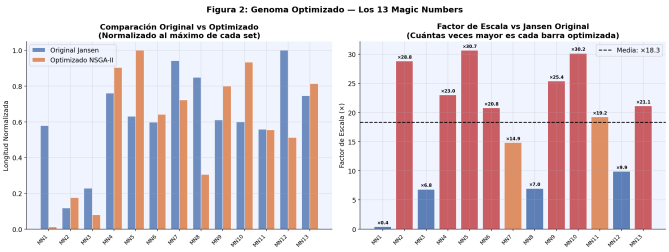


Fig. 3. Optimized LOKOGAD genome: the 13 Magic Numbers. Left: normalized comparison between original and optimized lengths. Right: per-bar scale factor (mean: $\times$18.3).

The links are assembled after printing, so the assembled model (approximately 40 cm) does not need to fit in the print bed. The genome corresponds to a single body size (that of the Gait2392 reference); adapting it to another patient would require regenerating the reference trajectory with a scaled OpenSim model and rerunning the pipeline, or uniformly scaling all bar lengths by the leg-length ratio, which preserves the trajectory shape (not evaluated here).

TABLE II
FINAL 12-BAR LOKOGAD MECHANISM: BAR LENGTHS

Bar	Nodes	Human scale (mm)	Prototype (mm)
0	0–2	76.70	13.1
1	2–3	491.55	84.2
2	1–3	423.03	72.5
3	1–4	393.19	67.4
4	2–4	493.06	84.5
5	3–5	546.87	93.7
6	1–5	241.89	41.4
7	5–6	377.57	64.7
8	6–7	537.07	92.0
9	4–7	667.19	114.3
10	2–6	640.75	109.8
11	1–2	85.94	14.7

TABLE III
FINAL FITNESS METRICS OF THE OPTIMIZED LOKOGAD EXOSKELETON

Metric	Value
Combined Fitness	0.11177
Foot Procrustes Error	0.04974 (4.97%)
Knee Procrustes Error	0.10791 (10.79%)
Absolute Position Error	0.21070 (21.07%)
Foot position error	0.0598 (5.98%)
Knee position error	0.4370 (43.70%)
Stride Length	966.35 mm
Valid Assembly	TRUE

C. Gait Sequence and Final Fitness Metrics

Fig. 4 presents the walking sequence along a 360° cycle across the 6 main gait phases, with an effective stride of 151.2 mm and vertical amplitude of 70.5 mm (55% stance, 45% swing), both at prototype scale. The *effective stride* (151.2 mm) is the horizontal extent of the foot during the stance region in the printed-scale model, whereas the *stride length* of 966.35 mm (Table III, Abstract) is the total horizontal foot excursion over the whole cycle at human scale; they are different quantities and are also expressed at different scales. Operationally, the stride length is $\max_\theta x_H(\theta) - \min_\theta x_H(\theta)$ over the 360 crank angles θ at human scale, where x_H is the horizontal coordinate of the foot point H; the effective stride is the same difference restricted to the angles for which the foot height satisfies $y_H \leq y_{min} + 0.35\,(y_{max} - y_{min})$, evaluated on the printed-scale model. Table III summarizes the winning individual's final metrics: a 4.97% foot error indicates a close match in trajectory shape with human gait, while the 10.79% knee error is an acceptable trade-off for a rigid-link mechanism without independent knee actuation. The absolute position error (21.07%) is not a shape metric: it measures how far the trajectory centroids lie from the reference ones relative to the hip. It is dominated by the knee (43.70%), while the foot centroid is well placed (5.98%). This means that the reproduced knee trajectory, although similar in shape, is displaced with respect to the human one; the accuracy claim is therefore limited to the foot trajectory.

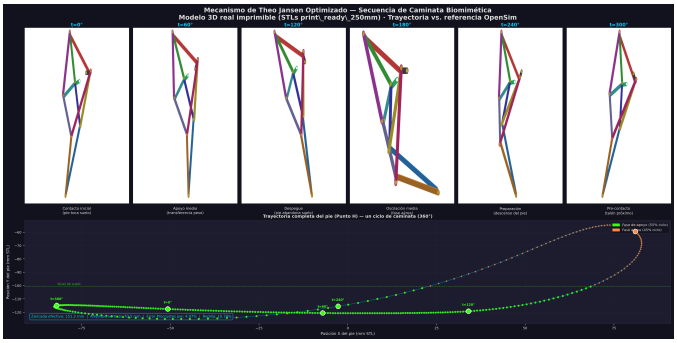


Fig. 4. Biomimetic gait sequence. Top: mechanism configurations at 6 phases ($t = 0°, 60°, 120°, 180°, 240°, 300°$). Bottom: full foot trajectory (point H), stance (55%) and swing (45%).

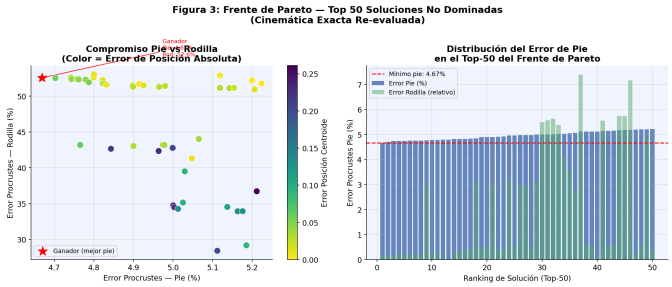


Fig. 5. Pareto front: Top-50 non-dominated solutions. Left: foot-vs-knee error trade-off (winning solution starred, foot 4.67%). Right: foot-error distribution.

D. Pareto Trade-offs and Z-Layer Collisions

The Pareto front reveals an inherent trade-off between foot- and knee-trajectory accuracy, reflecting the fundamental kinematic constraint of a rigid, single-DOF linkage (Fig. 5). This is particularly relevant for rehabilitation, since foot trajectory (stance and toe-off) is safety-critical while knee trajectory tolerates more variability. The 4.67% shown for the starred solution is its foot error in the Pareto-front evaluation, whereas the individual finally selected by the combined fitness (Table III) has a 4.97% foot error and a lower knee error; both values come from separate evaluation passes and are not contradictory. The optimized mechanism required 6 distinct Z-layers, with five bar pairs found in conflict across the full 360° cycle; total axial thickness is 33.0 mm, with the motor-coupling bar placed at the topmost layer for clearance from the motor housing (Fig. 6).

E. Physical Prototype

Fig. 7 shows the LOKOGAD prototype, comprising two Jansen mechanisms mounted on a bipedal support structure and driven by a single generic DC motor, with the two cranks offset by 180°. The prototype was built with common, inexpensive components (PLA/PETG filament, M3 hardware, standard ball bearings, and a generic motor). Since it is a proof of concept for the kinematics, no exact cost figure is claimed; what the architecture offers is that a single rotary input with a purely mechanical transmission removes the per-joint actuators, drivers, and sensors of multi-axis exoskeletons,

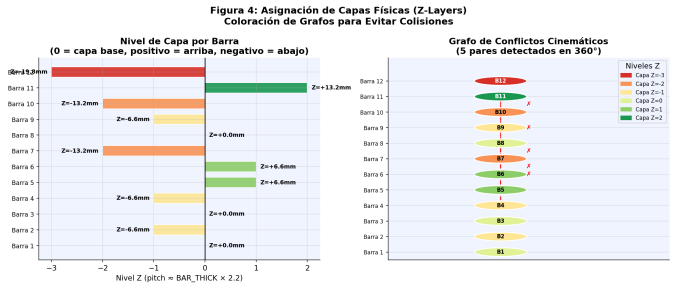


Fig. 6. Z-layer assignment via graph coloring (left) and kinematic conflict graph with 5 conflicting pairs (right).

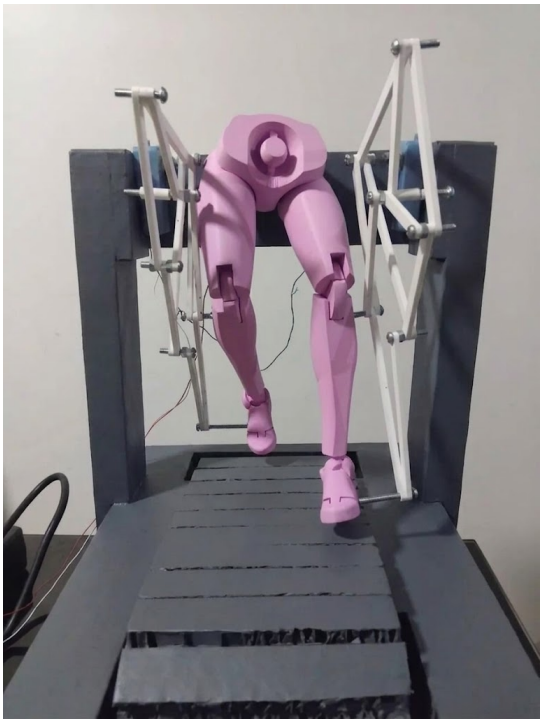


Fig. 7. Front view: Jansen limbs (white) coupled to biped support (magenta).

which in principle makes it less expensive. The prototype was not instrumented, so no trajectory was measured on it.

V. Conclusions

This paper presented LOKOGAD, a low-cost rehabilitation exoskeleton based on a biomimetic modification of the Theo Jansen gait mechanism for human bipedalism, designed via the GAD framework. The GAD paradigm successfully bridges computational graph theory and rehabilitation-exoskeleton design, enabling advanced evolutionary optimization to adapt existing locomotion mechanisms to human biomechanics. NSGA-II multi-objective optimization achieved a 4.97% Procrustes error in foot-trajectory replication against reference trajectories derived from the OpenSim Gait2392 model, a kinematic result obtained with a single-DOF mechanism. The exact analytical kinematic solver with robust NaN handling proved essential for maintaining geometric feasibility throughout evolution, removing the need for costly physical simulations and enabling efficient parallelization. The FDM-fabricated prototype, driven by a single generic DC motor, shows that the optimized design can be assembled and set in motion, and the single-DOF Jansen architecture reduces the actuator count and simplifies the control system, which in principle favors a lower-cost solution. These conclusions are limited to kinematics: forces, structural behavior, and tests with subjects were not addressed, and the prototype trajectory was not measured.

Future work includes measuring the prototype trajectories against the kinematic model, adding force/torque criteria and a structural analysis with a load-bearing frame, a cost analysis of a load-bearing version, integrating the URDF model with reinforcement-learning controllers (e.g., PPO, SAC) for patient-adaptive gait control; implementing foot-sole force sensors for load feedback during therapy; preliminary clinical trials to validate therapeutic effectiveness against conventional gait therapy; a portable version with a battery system and lightweight-alloy chassis for home use; and topological optimization beyond the Jansen architecture to improve knee-trajectory accuracy.

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